

Affordance-Grasp Conditioned Spill-Free Trajectory Optimization

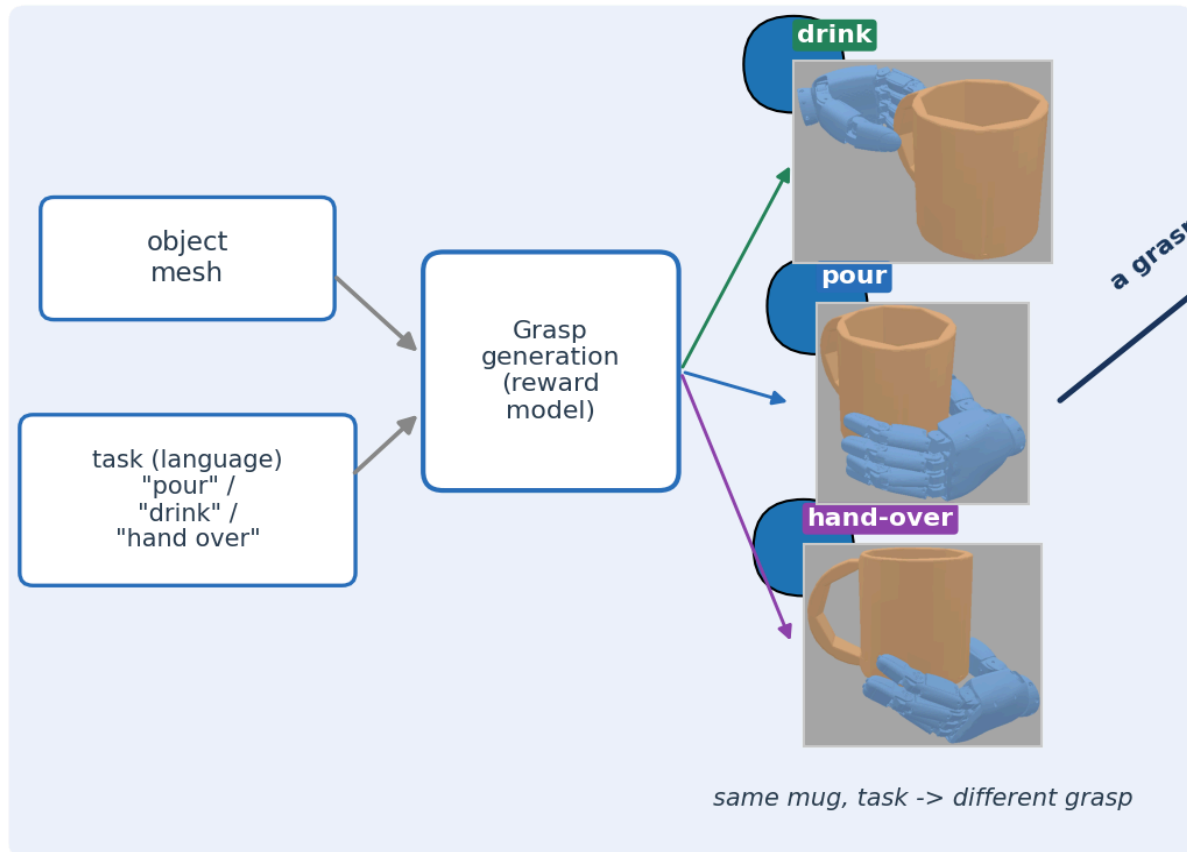
Does **how you grasp** a cup change **how safely you can carry** it?

Motion Planning · Proposal

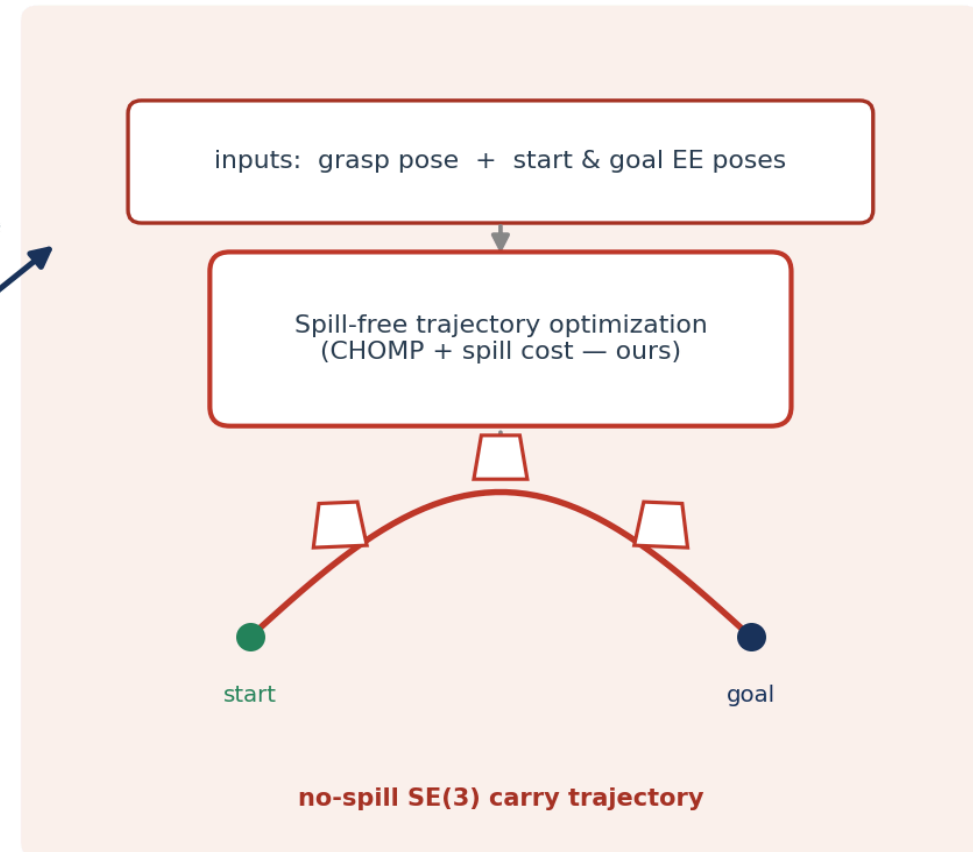
Jihoon Yun · 2026-06-08

Where this project fits — from grasp generation to the carry

MY THESIS · task-conditional grasp generation



THIS PROJECT · spill-free carry (motion planning)



My thesis (left): object mesh + a natural-language task → a dexterous grasp (same mug, the task picks a different grasp). **This project (right):** take that grasp + start/goal poses → plan a **spill-free** carry trajectory.

Problem — carry liquid from A to B without spilling

Given a grasped mug, plan a 6-DoF end-effector trajectory that is:

- **(a)** spill-free **(b)** obstacle-avoiding **(c)** smooth **(d)** tilt $\leq \theta_{max}$

Even a perfectly upright cup spills under a fast stop or turn — the danger is *motion*, not just orientation.

Why? That is the next slide.